PRASANTH RAYUDU

RTL Design Engineer: Verilog/SystemVerilog RTL Design, Verification, and Timing Closure

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PROFESSIONAL SUMMARY

- Computer Engineering graduate student and RTL Design Engineer with hands-on Verilog and SystemVerilog experience designing, integrating, and verifying digital logic blocks across FPGA and SoC subsystem projects.
- Designs control logic, datapaths, and execution units, including parallel MAC and ALU-style compute blocks, and resolves clock-domain crossing issues during SoC-level integration.
- Performs static timing analysis and achieves timing closure on Xilinx MPSoC FPGA prototypes using Vivado, ModelSim, and Xilinx ILA for waveform-level hardware debug.
- Builds reusable SystemVerilog testbenches, bus-functional models, and transactors, and applies UVM-based verification methods with coverage monitors to validate RTL correctness.
- Automates regression testing, benchmark generation, and simulation-to-emulation correlation with Python, Tcl, and Bash, reducing build cycles and debug iteration time through CI/CD integration.
- Graduate coursework in Computer Architecture and VLSI Design covers multi-core execution, cache-memory hierarchies, low-power microarchitecture, and digital logic fundamentals; communicates effectively during design reviews.

CORE SKILLS

RTL Design and Digital Logic: Verilog, SystemVerilog, VHDL, RTL design, control logic and datapath design, ALU/MAC-based execution units, finite state machines, low-power design techniques (clock gating)

Design Verification: UVM, testbench development, BFM/transactors, coverage monitors, functional verification, regression testing, pre-silicon validation, post-silicon bring-up correlation

Timing and Physical Implementation: Static timing analysis (STA), timing closure, clock-domain crossing (CDC) resolution, FPGA synthesis, Vivado place-and-route flows

Programming and Automation: Python, C, C++, Tcl, Bash/Shell, Make, CMake, Git, CI/CD, regression and benchmark automation, log parsing

Tools and Platforms: Synopsys VCS, ModelSim, Xilinx Vivado, Xilinx ILA, UltraScale/UltraScale+, MPSoC, Linux

Computer Architecture Foundations (graduate coursework): Multi-core and multi-threaded execution, cache-memory hierarchies, out-of-order execution concepts, low-power microarchitecture, VLSI design principles

Interfaces and Protocols: AXI, PCIe, I2C, LVDS, memory-mapped I/O, interrupts, ARM-based systems

WORK EXPERIENCE

Graduate Research Assistant, Hardware-Software Co-Design for Edge AI, San Diego State University Aug 2025 - Present

- Developed HyperOPT, a Python-based hardware-software co-design framework for architecture-level design-space exploration of accelerator configurations under accuracy, area, latency, and energy constraints.
- Modeled quantization, analog noise, and memory-cell non-idealities to quantify performance-per-watt and accuracy tradeoffs, verifying proxy estimates against detailed architecture-level results.
- Implemented pruning and binary-search algorithms to accelerate design-space exploration, producing reproducible benchmark reports for design reviews.

FPGA Emulation Flow Automation with CI/CD Integration Jan 2024 – aug2025

- Built a scalable FPGA emulation flow with CI-triggered builds, regression execution, and test generation on Xilinx UltraScale platforms for pre- and post-silicon verification.
- Developed Python, Tcl, Make, and Bash automation for compilation, environment configuration, and benchmark execution, integrating build and validation stages.
- Added log parsing and failure clustering enabling root-cause triage for over 95 percent of simulation-to-emulation mismatches, and reduced compile and debug iteration time by 30 percent.

Teaching Associate, General Studies, San Diego State University Aug 2024 - May 2025

- Led weekly first-year seminar sections, delivering instruction and timely assessment through Canvas LMS.
- Communicated complex material clearly to diverse audiences and collaborated in weekly teaching communities to improve outcomes.

Student Intern, FPGA and Machine Learning Systems, In22 Labs Jun 2021 - Jul 2022

- Designed and validated Verilog RTL blocks for a biomedical machine-learning pipeline, co-simulating embedded C reference models to verify functional correctness.

- Automated ModelSim waveform and log validation in Python, reducing manual debug effort by 50 percent and improving regression reproducibility.
- Collaborated with FPGA, software, and data-science engineers to reduce end-to-end inference latency and support ARM-based hardware-software integration.

SELECTED ENGINEERING PROJECTS

RTL Design and Verification of a Custom Execution Datapath Sep 2022 - Aug 2023

- Designed and verified RTL for a custom execution datapath, including control logic and ALU-style math datapath for texture and math engines, using Verilog and SystemVerilog.
- Resolved clock-domain crossing issues, conducted static timing analysis, and achieved timing closure on a Xilinx MPSoC FPGA prototype using Vivado.
- Collaborated with architecture and verification teams during design reviews to meet performance and power targets across compute workloads.

FPGA-Based Neural Network Video Processor Aug 2024 - May 2025

- Designed a Verilog datapath for a real-time inference pipeline built from parallel MAC-based ALU units, bias-addition logic, control FSMs, and ROM-based activation.
- Achieved 30 FPS and over 95 percent classification accuracy, performing RTL simulation, timing analysis, and on-board debug using Vivado, ModelSim, and Xilinx ILA.
- Evaluated throughput, resource utilization, and power tradeoffs while automating build and test-generation flows with Tcl and Make.

System-Level FPGA Emulation Framework for SoC Subsystem Validation Jan 2023 - Jun 2024

- Integrated a multi-IP CPU-microcontroller subsystem on a Xilinx MPSoC FPGA for pre-silicon validation and ARM embedded-Linux bring-up.
- Created reusable SystemVerilog testbenches, BFMs, and transactors for AXI, interrupts, PCIe, I2C, and LVDS interfaces, cutting build cycles 40 percent.

Dynamic Power Profiling Framework for SoC RTL Validation Jun 2023

- Developed a Python tool to parse VCD activity, compute switching factors, and estimate dynamic power across multiple RTL execution and datapath blocks.
- Generated IP-level reports on clock gating and bus utilization, showing 88 percent correlation to post-silicon measurements.

EDUCATION

Combined internship, research, and project experience since 2021 provides 3+ years of RTL design, verification, and digital design exposure.

Master of Science, Computer Engineering, San Diego State University (SDSU) Expected Dec 2026

Relevant Coursework: Computer Architecture, VLSI Design, Digital Systems, Embedded Systems

Bachelor of Technology, Electronics and Communication Engineering, VR Siddhartha Engineering College